

What is DOM?

DOM (Document Object Model) is a programming interface for HTML documents.

- It represents every HTML element as an **object**.
- JavaScript uses the DOM to **access, modify, add, or remove** HTML elements.
- Any change made using DOM is reflected immediately on the webpage without reloading it.

Basic DOM Flow

HTML Page



Browser Creates DOM



JavaScript Accesses DOM



Read / Change / Add / Delete Elements



Updated Webpage

Common DOM Methods

Method	Purpose
getElementById()	Select element using ID
getElementsByClassName()	Select elements by class
getElementsByTagName()	Select elements by tag
querySelector()	Select first matching element
querySelectorAll()	Select all matching elements
createElement()	Create a new HTML element
appendChild()	Add a new element
removeChild()	Remove an element

Common DOM Properties

Property	Purpose
innerHTML	Change HTML content
innerText	Change only text
textContent	Change text exactly as written
value	Read input value
style	Change CSS properties
src	Change image
className	Change class
id	Read or change ID

Common DOM Events

Event	Trigger
onclick	Mouse click
ondblclick	Double click
onmouseover	Mouse enters
onmouseout	Mouse leaves
onkeyup	Key released
onkeydown	Key pressed
onchange	Input changed
onsubmit	Form submitted

DOM Manipulation Process

Step 1

Select Element



Step 2

Perform Action



Step 3

Browser Updates HTML

Program 1

Change Text Content When Button is Clicked

Objective

Change the existing text displayed on the webpage after clicking a button.

DOM Concepts Used

- Selecting an element
- Handling button click event
- Changing text content

Logic

Button Click



Select Paragraph



Replace Old Text



Display New Text

Program 2

Change Background Color

Objective

Change the webpage background color using a button.

DOM Concepts Used

- Access body element
- Modify CSS through JavaScript

Logic

Button Click



Select Body



Change Background Color



Browser Refreshes Style

Program 3

Show and Hide Paragraph

Objective

Hide and display a paragraph using the same button.

DOM Concepts Used

- Style manipulation
- Conditional checking

Logic

Button Click

↓

Is Paragraph Visible?

↓

Yes

↓

Hide

OR

No

↓

Show

Program 4

Greeting User

Objective

Accept user's name and display a greeting message.

DOM Concepts Used

- Reading input value
- Displaying output

Logic

User Enters Name

↓

Read Input

↓

Create Greeting

↓

Display Message

Example Output

Enter Name

Prince

↓

Hello Prince!

Welcome.

Program 5

Change Image

Objective

Replace one image with another after button click.

DOM Concepts Used

- Access image element
- Modify image source

Logic

Button Click



Select Image



Replace Image Path



New Image Displayed

Program 6

Counter Application

Objective

Increase the displayed number every time the button is clicked.

DOM Concepts Used

- Variable
- Event handling
- Updating text

Logic

Initial Counter = 0



Button Click

↓

Counter + 1

↓

Update Screen

Example

0

↓

Click

↓

1

↓

Click

↓

2

↓

Click

↓

3

Program 7

Add New Item to List

Objective

Insert a new list item into an existing list.

DOM Concepts Used

- Create new element
- Append element

Logic

Button Click

↓

Create List Item

↓

Assign Text

↓

Append to List

↓

List Updated

Example

Before

Apple
Banana
After
Apple
Banana
Orange

Program 8

Remove Last Item From List

Objective

Delete the last item from the list.

DOM Concepts Used

- Select list
- Identify last child
- Remove element

Logic

Button Click

↓

Find Last Item

↓

Delete Last Item

↓

Updated List

Example

Before

Apple

Banana

Orange

After

Apple

Banana

Q1. What is DOM?

DOM is the Document Object Model that allows JavaScript to access and manipulate HTML elements.

Q2. Why is DOM used?

To make webpages interactive by changing content, style, structure, and responding to user actions.

Q3. What is DOM Manipulation?

Changing HTML elements using JavaScript.

Examples:

- Change text
 - Change color
 - Change image
 - Add elements
 - Remove elements
-

Q4. What are DOM Events?

Events are user actions detected by JavaScript.

Examples

- Click
- Double Click
- Keyboard
- Mouse
- Submit

Q5. Difference between innerHTML, innerText, and textContent

Property	Description
innerHTML	Reads or writes HTML content (including tags).
innerText	Reads or writes only the visible text.
textContent	Reads or writes all text exactly as stored, regardless of styling.

Q6. Difference between createElement() and appendChild()

Method	Purpose
createElement()	Creates a new HTML element
appendChild()	Adds the created element into the webpage

Q7. What is an Event?

An event is an action performed by the user or browser that JavaScript can respond to.

Examples:

- Click
- Key press
- Mouse hover
- Form submission